

Comment on “Multiobjective model predictive control” [Automatica 45 (2009) 2823–2830]

Víctor Valdivieso^a, Francisco G. Montoya^a, Alejandro Campa-Pinto^b,
Francisco Manuel Arrabal-Campos^{a,*}

^a*Department of Engineering, Research Centre CIAIMBITAL, University of Almería, 04120 Almería, Spain*

^b*Department of Computer Science and Electronics, Universidad de la Costa, Barranquilla 080002, Colombia*

Abstract

Lemma 4 of the commented paper states that the value function of the multi-parametric linear program with weight vector μ in the cost and state x in the right-hand side is convex and piecewise affine in μ for fixed x . The function is piecewise affine, but on its domain it is *concave* in μ , not convex: for each fixed decision variable the objective is affine in μ and the feasible set does not depend on μ , so the value function is a pointwise minimum of affine functions. We trace the slip to a duality remark in the companion conference paper, and we work out the consequences. The weight-selection linear program of Theorem 6 evaluates the value function as the *maximum* of its affine pieces and therefore imposes all of them at once; it describes a convex inner approximation of the true admissible weight set, which is in general non-convex. Every weight it returns satisfies the contractive constraint, so the closed-loop guarantees that rest on that constraint are preserved; but the claimed equivalence does not hold, and on randomly generated instances of the commented problem the program is strictly conservative in 97.5% of the levels tested and infeasible while admissible weights exist in 62.9% of them. In the quadratic case, the joint convexity invoked in Theorem 7 does not hold, because the bilinear coupling between weights and decision variables makes the joint Hessian indefinite unless that coupling van-

*Corresponding author.

Email address: fmarrabal@ual.es (Francisco Manuel Arrabal-Campos)

ishes; the online problem is not a convex program. Both defects admit an exact repair at the same cost as the published program: since the value function is the minimum of the affine pieces, the admissible weight set is their *union* of half-spaces, so it is obtained exactly by solving one linear program per active piece — which also restores the recursive feasibility witness that the published formulation loses.

Keywords: Predictive control, multiobjective optimization, multiparametric programming

1. The claim

The commented paper [1] builds a multiobjective MPC scheme in which, at each sample, a weight vector is selected online subject to the contractive constraint $V^*(\mu, x) \leq J_a$, where V^* is the optimal value of a multiparametric program with the weights in the cost. For the piecewise affine case, the machinery rests on [1, Lemma 4], which considers the multiparametric linear problem—their problem (13)—with parameters $\mu \in \mathbb{R}^l$ in the cost function and $x \in \mathbb{R}^n$ in the right-hand side of the constraints, and states that the value function V^* is

“continuous w.r.t. (μ, x) , convex and piecewise affine w.r.t. μ for any given x and w.r.t. x for any given μ .”

The statement for x is correct, and so is piecewise affinity in μ . The convexity claim for μ , however, has the wrong sign.

2. The correction

Lemma 1. *Let $V^*(\mu, x) = \min_z \{(c_0 + C'_\mu \mu)'z : Gz \leq b + Sx\}$ be the value function of problem (13) of [1]. Fix x such that the feasible set is nonempty, and let $D_x = \{\mu : V^*(\mu, x) > -\infty\}$ be the set of weights for which the program is bounded. Then $\mu \mapsto V^*(\mu, x)$ is concave on D_x , and piecewise affine there whenever the feasible set is pointed — as it is in the setting of the commented*

paper, where the decision variables are bounded below [2, Eq. (22)] and the input set is compact [1, footnote 3].

Proof. For each fixed feasible z the map $\mu \mapsto (c_0 + C'_\mu \mu)'z$ is affine, and the feasible set $\{z : Gz \leq b + Sx\}$ does not depend on μ . Hence $V^*(\cdot, x)$ is a pointwise infimum of a family of affine functions of μ , and therefore concave wherever it is finite. If the feasible set is pointed, the infimum is attained at one of its finitely many vertices, so on D_x the function is the minimum of finitely many affine functions and hence piecewise affine. $\square$

This is the classical sensitivity dichotomy of parametric optimization: under minimization the optimal value is convex in right-hand-side perturbations and concave in parameters that enter the objective linearly [3, Sec. 5.6.1, Ex. 3.9]. Both signs invert under maximization. The origin of the slip appears to be the companion conference paper [2, p. 2406], which establishes the properties in x and adds that “by duality, the same properties hold with respect to μ .” Duality does exchange right-hand-side and cost parameters, but it also turns the primal minimum into a dual maximum, and that is precisely the step that flips the curvature.

Numerical verification on the commented problem. We verified the corrected statement directly on problem (13), on randomly generated instances rather than on any particular controller, so that the evidence does not depend on how the scheme is instantiated. 200 random multiparametric linear programs with six decision variables, eight constraints and two weights, and six chord tests each (1,200 tests): the concavity inequality in μ held in every test, strictly in 791 of them and with equality in the rest, where the chord lay inside a single affine piece; the convexity inequality in μ held strictly in none. The same test in x — the direction the commented paper gets right — came out convex in every test, strictly in 320. A third test confirms the direction of the error at the level of the pieces: evaluating the value function as the *maximum* of the affine pieces, as the proof of Theorem 6 does, overstates it by up to 5.15 in value, while the

minimum reproduces it to 7×10^{-15} . Code and data are available with this comment.

3. Consequences within the commented paper

Theorem 6 and the weight-selection LP (17). The proof of Theorem 6 in [1] invokes the result of Schechter [5] for *convex* piecewise affine functions to evaluate $V^*(\mu, x)$, for fixed x , as the *maximum* of the affine pieces $\{(\phi_i x + \gamma_i)'(c'_0 + C'_\mu \mu)\}_{i \in I(x)}$, and the linear program (17) accordingly imposes

$$(\phi_i x + \gamma_i)'(c'_0 + C'_\mu \mu) \leq J_a \quad \text{for all } i \in I(x).$$

By Lemma 1 the value function is the *minimum* of those pieces. Requiring every piece to lie below J_a is therefore sufficient but not necessary for $V^*(\mu, x) \leq J_a$: the constraint set of (17) is a convex *inner approximation* of the true admissible weight set

$$\mathcal{A}(x, J_a) = \{\mu : V^*(\mu, x) \leq J_a\},$$

which, being a sublevel set of a concave function, is in general non-convex (for scalar μ it is a union of at most two intervals whenever the maximum of $V^*(\cdot, x)$ is interior and exceeds J_a).

Two things follow, one reassuring and one not. Since every weight accepted by (17) satisfies the contractive constraint, the closed-loop guarantees that rest on that constraint are untouched, and the piecewise affine scheme is safe as published. What fails is exactness. We quantified the gap on 480 instance-level pairs, sampling the weight simplex $\{\mu \geq 0, \sum_i \mu_i \leq 1\}$ declared in the commented paper: (17) was always contained in $\mathcal{A}(x, J_a)$, as the theory predicts, but *strictly* contained in 468 of them (97.5%); $\mathcal{A}(x, J_a)$ was non-convex in 232 (48.3%); and, most consequentially, (17) was *infeasible while admissible weights existed* in 302 of them (62.9%). In those samples the published program reports that no weight meets the contraction level when in fact a set of them does. Its optimizer, when feasible, also need not be the admissible weight closest to the desired one.

Theorem 7 and the quadratic case. For the branch with one quadratic objective, the proof of Theorem 7 in [1] asserts joint convexity of $V_\mu^*(\mu, x)$ in (μ, x) “by the results of Mangasarian and Rosen (1964),” and concludes that the online problem (21) is a convex program. In their problem (20) the weights enter the objective linearly through the bilinear term $\mu' C_\mu z$ while the constraints $Gz \leq b + Sx$ do not depend on μ , so the Hessian of the objective with respect to (z, μ) has the block form

$$\begin{bmatrix} H & C'_\mu \\ C_\mu & 0 \end{bmatrix},$$

which is positive semidefinite only if $C_\mu = 0$, that is, only if the weights do not act at all. The objective is therefore not jointly convex in (z, μ) , and partial minimization over z cannot be invoked to conclude joint convexity of the value function. Independently, the argument of Lemma 1 applies verbatim in the μ direction, so $V_\mu^*(\cdot, x)$ is concave; a function that is both concave and convex in μ is affine in μ , so joint convexity can hold only in that degenerate case. Consequently the feasible set of (21) is in general non-convex and (21) is not a convex program. Unlike the piecewise affine case, no inner-approximation safety net is created here; any candidate weight returned by a solver should therefore be certified a posteriori by evaluating V_μ^* directly, which is cheap.

4. An exact repair at the same cost

The correction also supplies the fix. Because $V^*(\cdot, x)$ is the *minimum* of the pieces, the admissible set is their *union* of half-spaces rather than their intersection:

$$\mathcal{A}(x, J_a) = \bigcup_{i \in I(x)} \{\mu \in \Delta : (\phi_i x + \gamma_i)'(c'_0 + C'_\mu \mu) \leq J_a\}, \quad (1)$$

with Δ the weight simplex. Each member of the union is defined by a *single* affine inequality over the same simplex, so selecting the admissible weight closest to a desired one is solved exactly by $\text{card}(I(x))$ linear programs of the same size as (17), and the best of their optima is returned. On the 480 instance-level pairs above, (1) reproduced the true admissible set exactly in 466 (97.1%;

the remaining discrepancies are grid points lying on the level itself), with the minimum of the pieces matching the value function to 2×10^{-9} .

The repair also restores a property the published program loses. The recursive feasibility argument of [1, Theorem 3] exhibits the previous weight as a witness: μ_{t-1} satisfies the contraction level by construction, hence $\mathcal{A}(x_t, J_a) \neq \emptyset$. That witness certifies the *minimum* of the pieces, which is exactly what (1) tests, whereas (17) tests their maximum — which is why (17) can be infeasible at a state where the scheme’s own witness is available. An implementation that keeps the previous weight whenever (17) is infeasible preserves the contractive constraint and therefore the guarantee; but with (1) the fallback is not needed, because the feasible set is never empty when the witness exists.

Gridding the simplex and testing $V^*(\mu, x) \leq J_a$ pointwise remains a practical alternative for very low weight dimensions, but it is an approximation, not an exact method, and we do not recommend it where (1) applies. Projection or gradient steps on the weight, and any construction that presumes convexity of $\mathcal{A}(x, J_a)$, can silently leave the admissible set.

Closing remark

The contribution of the commented paper—a multiobjective MPC formulation with closed-loop guarantees obtained through a contractive constraint on a weighted value function—is not in question, and the published piecewise affine algorithm remains safe precisely because the affected construction errs on the conservative side. The correction concerns the exactness claims attached to the weight-selection subproblems, and it comes with a repair that costs no more than the published program. The authors of the commented paper were contacted about this observation on 7 August 2026; no response had been received at the time of submission.

Data availability

The scripts and result files that generate every number reported here are distributed with this comment.

References

- [1] Bemporad, A., Muñoz de la Peña, D., 2009a. Multiobjective model predictive control. *Automatica* 45 (12), 2823–2830.
- [2] Bemporad, A., Muñoz de la Peña, D., 2009b. Multiobjective model predictive control based on convex piecewise affine costs. In: *Proceedings of the European Control Conference (ECC)*. Budapest, Hungary, pp. 2402–2407.
- [3] Boyd, S., Vandenberghe, L., 2004. *Convex Optimization*. Cambridge University Press, Cambridge, UK.
- [4] Mangasarian, O. L., Rosen, J. B., 1964. Inequalities for stochastic nonlinear programming problems. *Operations Research* 12 (1), 143–154.
- [5] Schechter, M., 1987. Polyhedral functions and multiparametric linear programming. *Journal of Optimization Theory and Applications* 53 (2), 269–280.